

# Asthma: Diagnosis and Management Guideline

Clinical guideline for variable respiratory symptoms, airflow limitation, controller therapy, and exacerbation prevention

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**Dataset status**

This document is a realistic synthetic artifact created to test document ingestion, chunking, retrieval, metadata filtering, reranking, citation, and question-answering workflows. It has not been reviewed or approved for patient care.

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**Controlled use**

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## 1. Introduction

Asthma is a heterogeneous chronic airway disease characterized by variable respiratory symptoms and variable expiratory airflow limitation. Airway inflammation, bronchial hyperresponsiveness, mucus, and reversible or fluctuating bronchoconstriction create symptoms that change over time and with triggers. [1-3]

Key considerations are: The guideline addresses diagnosis, assessment of control and future risk, inhaled corticosteroid-containing treatment, exacerbation management, difficult-to-treat asthma, and transitions between emergency and longitudinal care. A history of wheeze alone is insufficient for diagnosis; objective evidence of variable airflow limitation should be obtained whenever feasible. Regular use of a short-acting beta2 agonist without anti-inflammatory treatment is not an acceptable long-term strategy for most adolescents and adults.

Within the SVMH care pathway, Use this guideline with the approved asthma order sets, emergency pathways, medication formulary, and transition-of-care procedures. The Respiratory Care Quality Committee reviews safety events, delays, readmissions, and outcome measures related to asthma. Severe breathlessness, inability to speak, altered mental status, cyanosis, silent chest, exhaustion, marked airflow reduction, or poor response to initial bronchodilator therapy requires emergency escalation.

## 2. Purpose

This guideline standardizes screening, diagnosis, risk stratification, treatment, monitoring, consultation, and follow-up for asthma while preserving individualized clinical judgment.

Assessment should address the following: The guideline seeks to reduce preventable exacerbations, overreliance on reliever inhalers, inhaler-technique errors, and delayed recognition of severe or alternate disease. It emphasizes anti-inflammatory treatment, individualized action plans, trigger and comorbidity management, and objective reassessment before labeling asthma as treatment resistant. It promotes shared decision-making, clear escalation thresholds, and reliable communication between ambulatory, emergency, inpatient, diagnostic, pharmacy, and rehabilitation teams.

For routine implementation, Apply the guideline at initial presentation, during treatment change, after an acute event, and at each transition of care. Document the assessment, risk level, agreed plan, patient education, and timing of reassessment for asthma. Use current product labeling and specialty protocols when a recommendation depends on a device, medicine, procedure, or rapidly changing evidence base.

## 3. Scope

This guideline applies to adolescents and adults evaluated in ambulatory, emergency, observation, and inpatient settings. Pediatric patients below the local adolescent threshold use the pediatric asthma pathway.

The practical interpretation is as follows: Dedicated protocols govern status asthmaticus requiring critical care, procedural sedation, pregnancy-specific emergencies, and biologic-infusion reactions. The guideline applies to staff who diagnose asthma, prescribe or administer inhaled therapy, teach device technique, perform spirometry, or coordinate follow-up. People with occupational exposure, severe asthma, repeated steroid courses, or uncertain diagnosis may need specialty evaluation.

During assessment and treatment, Consult Pulmonary or Allergy for uncertain diagnosis, recurrent hospitalization, persistent poor control despite optimized therapy, suspected occupational asthma, biologic eligibility, or significant fixed airflow limitation. The document is reviewed at least every two years and sooner after a major guideline, regulatory, or safety update.

## 4. Anatomy, Physiology, and Background

Asthma involves variable narrowing of conducting airways, airway-wall inflammation, smooth-muscle hyperreactivity, mucus, and in some patients structural remodeling. Symptoms and airflow may normalize between episodes, which makes history and repeat objective testing important. [1,2]

Relevant clinical features include: Type 2 inflammation is common and may be associated with eosinophilia, allergic sensitization, elevated exhaled nitric oxide, or response to targeted biologic therapy. Non-type 2 patterns, obesity, smoking, infection, occupational exposure, and comorbid upper-airway disease can alter presentation and treatment response. Exacerbation risk is not captured fully by current symptoms; previous severe attacks, low lung function, excess reliever use, and poor inhaled corticosteroid adherence increase future risk.

At each transition of care, Assess both symptom control and future exacerbation risk at every review rather than using a symptom score alone. Problem-list terminology should identify the clinically meaningful subtype, severity, stage, and major complications when known. Explain the disease mechanism in language that supports the patient or caregiver to recognize symptoms and understand the treatment rationale.

Table 1. Asthma control domains

| Domain               | Examples                                                            | Why it matters                                             |
|----------------------|---------------------------------------------------------------------|------------------------------------------------------------|
| Current symptoms     | Day symptoms, night waking, activity limitation, reliever use       | Guides current control assessment and treatment discussion |
| Future risk          | Prior severe attack, low FEV1, reliever overuse, poor ICS adherence | Predicts exacerbation even when symptoms seem mild         |
| Treatment factors    | Technique, adherence, device, access, side effects                  | Common reversible causes of poor control                   |
| Comorbidity/exposure | Rhinitis, reflux, obesity, smoking, workplace sensitizers           | May mimic or worsen asthma and alter response              |

## 5. Causes and Etiologic Classification

Asthma reflects genetic susceptibility interacting with allergens, viral infection, irritants, occupational agents, exercise, weather, medicines, and other exposures. A single trigger should not be assumed to be the sole cause.

The core elements are: Common triggers include respiratory viral infections, allergens, smoke, air pollution, strong odors, cold air, exercise, and emotional stress. Aspirin and other cyclooxygenase-1 inhibitors can provoke respiratory reactions in susceptible patients, and nonselective beta blockers may worsen bronchospasm. Work-related asthma may improve away from the workplace and requires a careful exposure timeline and objective testing.

When the guideline is applied in practice, Ask about home, school, workplace, hobbies, pets, mold, tobacco or vaping, medicines, and temporal patterns before recommending avoidance measures. Record the evidence supporting the working classification and update it when subsequent testing changes the diagnosis.

6. Risk Factors

Risk factors for asthma development or poor outcomes include atopy, family history, tobacco exposure, occupational sensitizers, obesity, rhinitis, recurrent viral illness, socioeconomic barriers, and prior severe attacks.

Key considerations are: A previous intubation, intensive-care admission, or severe exacerbation is a major marker of future risk. High short-acting beta2-agonist use, no inhaled corticosteroid, poor technique, low lung function, and repeated oral corticosteroid courses indicate unsafe control. Mental-health conditions, low health literacy, cost, fragmented care, and inability to obtain inhalers or spacers can increase preventable emergency use.

Within the SVMH care pathway, Review reliever refill patterns, controller access, inhaler technique, and the written action plan rather than relying only on patient recall. Quality reports should be stratified when feasible to identify inequities in screening, treatment access, and outcomes.

- Record prior emergency visits, hospitalizations, intensive care, and oral steroid courses.
- Check smoking, vaping, and occupational exposure at follow-up.
- Treat modifiable risks even when day-to-day symptoms are infrequent.

7. Symptoms and Clinical Presentation

Typical symptoms are episodic wheeze, shortness of breath, chest tightness, and cough that vary in intensity and occur in characteristic patterns. Nocturnal symptoms, exercise symptoms, and worsening with viral infection or exposure are common.

Assessment should address the following: Asthma can present primarily as cough, but isolated chronic cough also has many alternate causes. Physical examination may be normal between episodes; wheeze can be absent in a life-threatening attack when airflow is critically reduced. Signs of severe exacerbation include inability to complete sentences, agitation or drowsiness, accessory-muscle use, low oxygen saturation, marked tachypnea, and poor air movement.

For routine implementation, Measure oxygen saturation and objective airflow when possible, but do not delay initial treatment in a clinically severe attack. Time-sensitive findings require direct clinician-to-clinician escalation rather than passive electronic messaging. Give rapid bronchodilator and systemic corticosteroid treatment under the acute pathway, reassess frequently, and escalate when response is incomplete or deterioration is present.

Warning

A quiet or silent chest in a distressed patient is a danger sign, not evidence that wheezing has improved.

8. Diagnosis

Diagnosis requires a compatible history plus evidence of variable expiratory airflow limitation, ideally before long-term controller treatment is started. If treatment has already begun, confirmation may require supervised step-down or repeat testing when safe. [1,2]

The practical interpretation is as follows: Spirometry with bronchodilator response is the preferred initial objective test in most adolescents and adults. Peak-flow variability, improvement after anti-inflammatory treatment, exercise or bronchial challenge, and between-visit variability can provide additional evidence. Exhaled nitric oxide and eosinophils can support a type 2 inflammatory phenotype but do not confirm asthma by themselves and can be affected by treatment and comorbidity.

During assessment and treatment, Repeat objective testing when initial spirometry is normal but clinical suspicion remains, and document the conditions under which testing was performed. The record should state the diagnostic basis, severity, uncertainty, and plan for confirmation or reassessment.

Table 2. Diagnostic evidence for variable airflow limitation

| Method               | Supportive pattern                                                      | Important limitation                                                    |
|----------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Spirometry           | Obstruction with clinically meaningful improvement after bronchodilator | May be normal between episodes or altered by prior controller treatment |
| Peak-flow diary      | Excessive day-to-day variability or improvement with treatment          | Technique and device consistency are essential                          |
| Challenge testing    | Airway hyperresponsiveness under a validated protocol                   | Requires appropriate setting and medication preparation                 |
| Inflammatory markers | Elevated FeNO or eosinophils consistent with type 2 inflammation        | Supports phenotype; not diagnostic alone                                |

9. Laboratory Tests

Laboratory testing is selective and is used to evaluate phenotype, alternate diagnoses, complications, and treatment safety rather than to diagnose routine asthma.

Relevant clinical features include: Complete blood count with eosinophils, total or specific IgE, and allergy testing may support phenotype and trigger assessment in selected patients. Arterial or venous blood gas is reserved for severe attacks, fatigue, altered mental status, or concern for ventilatory failure. Electrolytes and glucose may be relevant during repeated beta-agonist treatment, systemic corticosteroid use, or critical illness.

At each transition of care, Interpret eosinophils and exhaled nitric oxide in relation to recent corticosteroids, smoking, infection, and adherence. Avoid routine repeat testing that is unlikely to change treatment, and use critical-result communication procedures when thresholds are met.

Table 3. Selective laboratory use

| Test                     | Potential use                                                   | Do not use as                                   |
|--------------------------|-----------------------------------------------------------------|-------------------------------------------------|
| Blood eosinophils        | Phenotype, biologic eligibility, alternate eosinophilic disease | A stand-alone asthma diagnostic test            |
| Specific IgE / skin test | Identify clinically relevant sensitization                      | Proof that an exposure is causing every symptom |
| Blood gas                | Assess ventilatory failure in severe exacerbation               | Routine testing in a mild attack                |
| Electrolytes/glucose     | Monitor high-dose beta agonist or systemic steroid effects      | Routine chronic control assessment              |

10. Imaging and Additional Diagnostics

Chest imaging is not routine for uncomplicated asthma but is appropriate when the presentation is atypical, severe, focal, recurrent in one location, or poorly responsive to expected treatment.

The core elements are: Chest radiography may identify pneumonia, pneumothorax, heart failure, foreign body, mass, or other alternate disease. Computed tomography is reserved for selected questions such as bronchiectasis, allergic bronchopulmonary aspergillosis, severe asthma phenotype, or another structural disorder. Laryngoscopy, sleep testing, cardiac evaluation, or swallowing studies may be appropriate when symptoms suggest an alternate or coexisting condition.

When the guideline is applied in practice, Avoid repeated imaging for typical exacerbations without red flags, and use radiation-conscious protocols when imaging is required. Incidental and urgent findings require closed-loop communication and a named clinician responsible for follow-up. Orders should include a focused clinical question and the information needed to select the safest protocol.

11. Differential Diagnosis

Many disorders mimic or coexist with asthma, especially when symptoms are persistent, spirometry is normal, or expected treatment does not work.

Key considerations are: Important alternatives include inducible laryngeal obstruction, chronic obstructive pulmonary disease, bronchiectasis, heart failure, pulmonary embolism, dysfunctional breathing, reflux, upper-airway cough syndrome, and medication-related cough. Foreign body, central airway obstruction, infection, eosinophilic lung disease, and occupational disease should be considered when the pattern is atypical. Anxiety can amplify breathlessness but should not be used to dismiss objective respiratory symptoms or delay evaluation.

Within the SVMH care pathway, Reconsider the diagnosis before escalating to prolonged systemic corticosteroids or advanced biologic therapy. Discharge documentation should distinguish confirmed disease, suspected disease pending confirmation, and alternate diagnoses that require follow-up.

## 12. Treatment

Long-term treatment uses an inhaled corticosteroid-containing regimen for anti-inflammatory protection, with stepwise adjustment according to symptoms, exacerbation risk, lung function, adherence, technique, and preference. [1-4]

Assessment should address the following: For many adolescents and adults, an inhaled corticosteroid-formoterol reliever strategy reduces severe exacerbations compared with short-acting bronchodilator-only treatment. Before stepping up, confirm the diagnosis, inhaler technique, adherence, exposure, and comorbidity; after sustained control, consider cautious step-down while maintaining inhaled corticosteroid protection. Every patient should receive a written action plan describing daily treatment, worsening symptoms, reliever use, and when to seek urgent care.

For routine implementation, Demonstrate and observe inhaler technique using the actual device, and match the device to inspiratory ability, dexterity, cognition, and coverage. Treatment delays, contraindications, and patient preferences should be documented with the risk-mitigation plan.

**Practice note**  
Do not stop inhaled corticosteroid completely in most adolescents and adults with confirmed asthma unless temporary cessation is required for supervised diagnostic confirmation.

## 13. Medications

Asthma medicines include inhaled corticosteroids, inhaled corticosteroid-formoterol combinations, other long-acting bronchodilators used with corticosteroid, leukotriene receptor antagonists, long-acting muscarinic antagonists, systemic corticosteroids for exacerbations, and phenotype-directed biologics.

The practical interpretation is as follows: Short-acting beta2 agonists relieve bronchoconstriction but do not treat airway inflammation and should not be the only ongoing therapy for most patients. Oral corticosteroid courses should be limited to appropriate exacerbations because cumulative exposure increases metabolic, bone, eye, infection, and psychiatric risk. Biologic selection depends on severe-asthma confirmation, phenotype, exacerbation history, biomarkers, comorbidity, prior optimized therapy, and access.

During assessment and treatment, Review dose, device, spacer use, refill history, oral steroid exposure, side effects, and whether the reliever and controller plan are understood. High-alert medicines and infusions require the safeguards in the Medication Administration SOP and current pharmacy guidance. Reconcile what the patient is actually taking, including nonprescription products, dose timing, adherence barriers, and recent changes.

Table 4. Asthma medication roles

| Therapy                 | Primary role                                                 | Key safety point                                                    | Monitoring focus                                      |
|-------------------------|--------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------|
| ICS-containing inhaler  | Foundation of anti-inflammatory treatment                    | Rinse mouth when appropriate; verify technique and adherence        | Symptoms, exacerbations, lung function, local effects |
| ICS-formoterol reliever | Relief plus anti-inflammatory treatment in eligible regimens | Use only with an approved formulation and clear action plan         | Use frequency and response                            |
| SABA                    | Rapid bronchodilation when prescribed                        | Overuse signals risk; not adequate as sole long-term therapy        | Canister/refill use and symptoms                      |
| Biologic                | Add-on for selected severe phenotype                         | Screening, administration, and response criteria are agent-specific | Exacerbations, steroid reduction, adverse effects     |

## 14. Contraindications and Precautions

Precautions include medicine interactions, incorrect device use, excessive reliever use, cumulative systemic corticosteroid exposure, and conditions that make a presumed asthma diagnosis unsafe.

Relevant clinical features include: Long-acting beta2 agonist therapy should not be used without inhaled corticosteroid treatment in asthma. Nonselective beta blockers and aspirin or NSAIDs can provoke symptoms in susceptible patients and require individualized risk assessment. Sedatives may worsen ventilatory failure in a severe attack and should not be used to manage respiratory distress.

At each transition of care, During procedures or pregnancy, continue necessary controller therapy unless a responsible clinician provides a specific alternative plan. A prior reaction should be described precisely rather than recorded as an unexplained class allergy.

**Warning**

Do not delay escalation or intubation planning because the patient has a normal carbon dioxide value; a rising or normalizing carbon dioxide in a tiring patient can indicate impending respiratory failure.

**15. Complications**

Poorly controlled asthma causes recurrent exacerbations, emergency visits, hospitalization, loss of function, fixed airflow limitation, and adverse effects from repeated systemic corticosteroid treatment.

The core elements are: Severe attacks can cause hypoxemia, pneumothorax, respiratory failure, cardiac arrest, and death. Chronic complications include sleep disruption, school or work absence, anxiety, activity avoidance, and treatment burden. Medication complications include oral candidiasis, dysphonia, tremor, tachycardia, adrenal suppression, osteoporosis, cataract, and metabolic effects depending on exposure.

When the guideline is applied in practice, After every exacerbation, review the trigger, controller plan, inhaler technique, adherence, action plan, and follow-up within an appropriate interval. Relevant complications, severity, referral status, and closure of the follow-up loop should be visible in the record.

**16. Lifestyle Recommendations**

Lifestyle management focuses on avoiding harmful exposure without creating unnecessary restriction. Physical activity is encouraged when asthma is controlled and pre-exercise treatment is individualized.

Key considerations are: Eliminate tobacco and vaping exposure, reduce clinically relevant allergens or workplace sensitizers, and improve ventilation and moisture control when feasible. Treat obesity, rhinitis, sleep apnea, and other comorbidities that worsen symptoms or complicate assessment. Vaccination, hand hygiene, sleep, stress support, and a plan for viral respiratory illness reduce disruption and risk.

Within the SVMH care pathway, Avoid expensive environmental interventions unless the exposure is clinically relevant and the proposed measure has a plausible benefit. Offer multidisciplinary support when nutrition, activity, sleep, tobacco, alcohol, stress, or social circumstances materially affect risk. Select one or two realistic goals, identify barriers, and arrange review instead of providing an unprioritized list.

- Warm up before exercise and carry the prescribed reliever.
- Use a spacer with a pressurized metered-dose inhaler when indicated.
- Check air-quality and workplace exposure patterns when symptoms fluctuate by location.

**17. Follow-up Care**

Follow-up assesses symptom control, exacerbation risk, lung function, treatment burden, technique, adherence, and the action plan. Review is especially important after an emergency visit or oral corticosteroid course.

Assessment should address the following: Repeat spirometry periodically and after meaningful treatment change when results will inform decisions. Ask about nighttime symptoms, activity limitation, reliever use, missed doses, oral steroid exposure, and adverse effects. Step-down is considered only after sustained control with a clear follow-up and action plan.

For routine implementation, Arrange early post-exacerbation review and ensure the patient leaves acute care with an inhaled corticosteroid-containing plan and access to medicines and devices. Missed follow-up after a high-risk event should trigger outreach under the Transitions of Care procedure.

**Table 5. Asthma follow-up domains**

| Domain      | Review focus                                                   | Escalate when                                               |
|-------------|----------------------------------------------------------------|-------------------------------------------------------------|
| Control     | Symptoms, night waking, activity, reliever use                 | Rapid worsening, frequent reliever, repeated night symptoms |
| Risk        | Prior attacks, lung function, steroid courses, refill pattern  | Hospitalization, ICU history, low FEV1, recurrent steroids  |
| Treatment   | Technique, adherence, device fit, side effects, cost           | Unable to obtain controller or repeated technique failure   |
| Comorbidity | Rhinitis, reflux, obesity, sleep, smoking, vocal cord symptoms | Alternate diagnosis or severe phenotype suspected           |



## 18. Special Populations

Pregnancy, adolescence, older age, smoking-related disease, and occupational exposure require tailored assessment while maintaining effective anti-inflammatory treatment.

The practical interpretation is as follows: In pregnancy, poor asthma control poses risk to both pregnant patient and fetus; necessary inhaled corticosteroid-containing treatment is generally continued under obstetric coordination. Adolescents need confidential assessment of adherence, vaping, mental health, school access, and transition to adult care. Older adults may have overlapping chronic obstructive pulmonary disease, cardiac disease, reduced inspiratory flow, and greater device or polypharmacy challenges.

During assessment and treatment, Provide school, workplace, travel, and emergency documentation when access to inhalers or accommodations depends on it. Consultation thresholds should be lower when communication, frailty, pregnancy, organ dysfunction, or caregiver limitations increase risk.

## 19. Prevention

Exacerbation prevention depends on inhaled corticosteroid-containing therapy, correct technique, adherence, trigger and comorbidity management, vaccination, and a usable written action plan. [1,3]

Relevant clinical features include: Monitor reliever overuse and repeated oral corticosteroid courses as active safety signals. Teach patients to respond early to worsening symptoms and to seek urgent care for severe or rapidly progressive signs. Reduce tobacco, vaping, occupational, and indoor air exposures through practical interventions and supportive policy.

At each transition of care, Use pharmacy refill data and post-exacerbation outreach to identify preventable gaps. Prevention measures should be tracked as completed, not merely ordered, and abnormal findings require documented action. Use community resources, remote monitoring, pharmacist support, and care management when they improve access and continuity.

## 20. Quality Indicators and Audit

Asthma quality measures should emphasize severe-exacerbation prevention, inhaled corticosteroid access, technique, action plans, objective diagnosis, and equitable follow-up.

The core elements are: Track emergency revisits, hospitalization, intensive-care admission, oral corticosteroid bursts, and high reliever use. Audit whether objective diagnostic testing, inhaler-technique observation, and post-exacerbation follow-up were completed. Stratify controller access and outcomes by age, language, insurance, and care setting to detect barriers.

When the guideline is applied in practice, The Respiratory Care Quality Committee reviews trends quarterly and tests interventions that reduce preventable attacks and steroid exposure. Audit results are reported to the Respiratory Care Quality Committee and the Medical Executive Committee with assigned actions and completion dates.

Table 6. Example asthma quality indicators

| Indicator                     | Operational definition                                                             | Frequency |
|-------------------------------|------------------------------------------------------------------------------------|-----------|
| ICS-containing discharge plan | Eligible acute-care discharges with documented controller/reliever plan and access | Monthly   |
| Technique verification        | Visits with observed inhaler technique and correction when needed                  | Quarterly |
| Post-exacerbation review      | Completed follow-up after emergency visit, hospitalization, or steroid burst       | Monthly   |
| Reliever overuse review       | Patients with high reliever dispensing who received clinical outreach              | Quarterly |

## 21. References

References are provided to make the synthetic document realistic and to support citation and provenance tests. They should be checked against the most current source before any real-world use.

[1] Global Initiative for Asthma. Global Strategy for Asthma Management and Prevention. 2026 update.

[2] Global Initiative for Asthma. Difficult-to-Treat and Severe Asthma in Adolescent and Adult Patients: Diagnosis and Management. 2026.

[3] National Asthma Education and Prevention Program Coordinating Committee. 2020 Focused Updates to the Asthma Management Guidelines.

[4] U.S. Food and Drug Administration. Current prescribing information for inhaled and biologic asthma therapies.

**Reference status**

External guidelines and regulations may change between scheduled reviews. The document owner is responsible for checking high-impact updates and initiating an interim revision when necessary.